

Parallel Scenario

Question: What is the average water consumption in liters of the top 3 SP500 companies in 2023, considering only UK branches?

Extract one value from each non-relational table, then aggregate.

Company X			Company Y			Company Y		
2023 indicators per employee	Branch location		Branch locations 2023	Weekly liters (per employee)		Weekly usage per employee	Rbnach citise	
	UK	US		Water	Oil		Lodnon (UK)	Tokoy (Jaapn)
Weekly water usage 2022	5.89 gallons	7.16 gallons	France	22	31	Waetr	0.022 m^3	0.031 m^3
Weekly water usage 2023	5.60 gallons	--	UK	21.5	not available		0.024 m^3	0.028 m^3
Perturbations: Pivot + unit change + n.a.			Perturbations: Pivot + unit in header + n.a.			Perturbations: Row/Col pivot + unit change + OCR/typos		

Conversion: 5.6 gallons $\rightarrow$ 21.2 liters, 0.024 $m^3 \rightarrow$ 24 liters
 avg(21.2, 21.5, 24)

Answer:
22.23 liters

Sequential Scenario

Question: What is the average 2023 investment amount of the main lab of the company where John Doe is CEO?
 Restrict the calculation to investments done in AI and the financial sector

Follow implicit lookup keys across perturbed tables.

CEO names			Facility assignment				R&D expenses		
Year	Compaines		Facilities	Companies			Costs 2023	Investments (millions)	
	X	Y		X	Y			AI Research	Financial Research
2022	John Doe	Lee Seo-yeon	Primary	Atlanta Lab	Seoul Lab		Net cost	Atlanta Lab	42.1
2023	Jnae Mya	n.a.	Backup	Denver Hub	Busan Hub			Seoul Lab	53.8 - as written in [1]
Perturbations: Row/Col pivot + n.a. + OCR/typos			Perturbations: Pivot + empty column				Perturbations: Row/Col pivot + text insertion		

propagate values
 Company X $\rightarrow$ Atlanta Lab $\rightarrow$ avg(42.1, 38.9)

Answer:
40.5